INCH-POUND
MIL-DTL-32024(OS)
14 May 1998
SUPERSEDING
AS 3009B
12 January 1994

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, CCU-33/A

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the CCU-33/A Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or documents recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements for documents referenced in sections 3 and 4 of this specification, whether or not they are listed in this section.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-P-20444 Primer, Percussion, M42 Parts for Loading, Assembling, and Packing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Team (Code 840N), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377
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MIL-PRF-81352 Coating, Aircraft Touch-Up

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810C Environmental Test Methods and Engineering Guidelines
DOD-STD-2101 Classification of Characteristics

(Unless otherwise indicated, copies of Federal and Military specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL825AS100 Cartridge, Impulse, CCU-33/A Assembly
838AS102 Powder, Smokeless
838AS144 Test Apparatus
844AS291 Vibration Fixture for JAU-22/B Initiator

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Data Control Team (Code 8410P), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742 Standard Practice for Radiographic Examination

(Application for copies of ASTMs should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN SOCIETY FOR QUALITY CONTROL

ANSI/ASQC Z1.4 Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Ave., Milwaukee, WI 53201-3005.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referenced on the respective Naval Air Systems Command drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant field activity for cartridges prior to manufacture. When alternate materials or methods of cartridge manufacture are specified on the drawings, the bidder's selection shall be clearly stated in the proposal.

3.3 Primary components. For the purpose of this specification, the percussion primer, conforming to MIL-P-20444, and the smokeless powder, conforming to Drawing 838AS102, are defined as primary components (see 3.3.1 and 3.3.2).

3.3.1 Percussion primers. The M42C1-PA101 percussion primers used in the cartridge shall be manufactured and tested in accordance with MIL-P-20444. The primers shall be loaded into the cartridges within 24 months of new primer acceptance testing or within 12 months of primer recertification testing.

3.3.2 Smokeless powder. The smokeless powder shall be in accordance with Drawings 825AS100 and 838AS102 and loaded as prescribed therein. Smokeless powder charges shall be determined by weight rather than by volume measurements.

3.3.2.1 Humidity. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity of the working chamber to exceed 75 percent at ambient temperature to avoid undesirable effects upon ballistic performance. Testing shall be in accordance with 4.9.

3.4 Visual inspection.

3.4.1 Surface characteristics. The cartridge shall contain no dents, deep scratches, burred threads, longitudinal cracks in the crimped areas, lose crimps, or any other defect which may affect proper installation, stability, or moisture resistance of the cartridge when tested in accordance with 4.7.2. Fissures in the crimped area are acceptable if coated with acrylic lacquer conforming to Type I of MIL-PRF-81352. Circumferential cracks in the crimped area less than 0.125-inch in length and more than 0.375 inch apart are acceptable if coated with acrylic lacquer conforming to Type I of MIL-PRF-81352. Loose crimps, cracks, and fissures are defined in 6.4.2, 6.4.3, and 6.4.4, respectively.

3.4.1.1 Thread gage. The cartridge shall be accepted by passing a thread gage test for 1.000-16UN-2A threads.

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3.5 Radiographic examination. Cartridges shall be free of imperfections in components of assembly as verified by radiographic examination of all first article and production test samples (see 4.7.3).

3.6 Leakage. The cartridge shall not exhibit a leak rate in excess of 1×10^{-5} cubic centimeters per second (cm^3/sec) of gas at a pressure differential of 1.0 ± 0.1 atmosphere when tested in accordance with 4.7.4.

3.7 Environmental tests.

3.7.1 Shock. The cartridges shall withstand shock conditions as specified in 4.2.2.1. After being subjected to the shock test, the cartridges shall meet the nondestructive test requirements of 3.5 and 3.6, and the design performance requirements of 3.8 and 3.9 when test fired.

3.7.2 Vibration. The cartridges shall withstand vibration conditions as specified in 4.2.2.2. After being subjected to the vibration test, the cartridges shall meet the nondestructive test requirements of 3.5 and 3.6, and the design performance requirements of 3.8 and 3.9 when test fired.

3.7.3 Temperature/humidity/altitude cycling. The cartridges shall withstand the temperature/humidity/ altitude cycling conditions as specified in 4.2.2.3. After being subjected to the temperature/humidity/ altitude cycling test, the cartridges shall meet the nondestructive test requirements of 3.5 and 3.6, and the design performance requirements of 3.8 and 3.9 when test fired.

3.8 Functional.

3.8.1 Low temperature. The cartridge shall withstand the functional requirement of -65°F when tested in accordance with 4.2.3.1.

3.8.2 Normal temperature. The cartridge shall withstand the functional requirement of 70°F when tested in accordance with 4.2.3.2.

3.8.3 High temperature. The cartridge shall withstand the functional requirement of 160°F when tested in accordance with 4.2.3.3.

3.9 Ballistics.

3.9.1 Velocity. The velocity of ballistic test slugs shall have an upper limit of 225 feet per second (ft/sec) and a lower limit of 175 ft/sec when tested in accordance with 4.8.

3.9.2 Functioning time. The cartridge functioning time shall be less than 25 milliseconds when tested in accordance with 4.8 (see 6.4.1).

3.9.3 Misfire. The cartridge shall fire when proper firing energy is applied (see 4.8.3).

3.10 Packaging and marking. The packaging of cartridges and the container markings shall be accordance with section 5 and as specified in the contract (see 6.2).

3.11 Workmanship. The cartridge assembly shall be manufactured and finished in a manner that assures

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compliance with all requirements of this specification. No interpretation of any part of this or other applicable documents shall be construed to relieve the contractor of his responsibility to furnish the cartridge with the required degree of quality and workmanship.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Production inspection (see 4.3)

4.2 First article inspection. Before entering into quantity production, a first article sample of 66 cartridges shall be prepared (see table I). Additional cartridges, as specified in the contract, shall be retained for investigative purposes. A first article sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of 3.2 through 3.6 and 3.11 of this specification. The failure of any cartridge to meet the acceptance requirements as stated above shall cause rejection of the first article sample. Any quantity production prior to notification by the contracting activity of first article sample acceptability shall be at the contractor's risk.

TABLE I. First article inspection.

Nature of requirement	Number of samples							Para.
Visual inspection	6	6	18	12	12	12	*	4.7.2
Radiographic examination	6	6	18	12	12	12	*	4.7.3
Leakage	6	6	18	12	12	12	*	4.7.4
20g shock	6							4.2.2.1
Vibration		6						4.2.2.2
Temperature/humidity/altitude			18					4.2.2.3
Ballistic test at: -65 ± 5°F	2	2	6	12				4.2.3.1 & 4.8
70 ± 5°F	2	2	6		12			4.2.3.2 & 4.8
160 ± 5°F	2	2	6			12		4.2.3.3 & 4.8

* Retain samples, see contract

4.2.1 First article sampling. First article sampling procedures and rejection criteria, except for the environmental tests of 4.2.2 and the functional tests of 4.2.3, shall be in accordance with the requirements of ANSI/ASQC Z1.4. Critical, major, and minor characteristics shall be inspected for defects to acceptable quality levels (AQLs) as specified in the contract (see 6.2 and 6.6). The requirements of ANSI/ASQC Z1.4 pertaining to normal and tightened inspections shall apply. A production lot of cartridges shall contain main propellant charges and primers from single production lots only.

4.2.2 Environmental test sampling.

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4.2.2.1 Shock. This test shall be conducted in accordance with MIL-STD-810C, Method 516.2, Procedure I, and Figure 516.2-1. The shock pulse wave form shall be terminal peak sawtooth. The peak amplitude shall be 20g and the duration shall be 11 milliseconds. After the shock test has been completed, the cartridges shall be subjected to the nondestructive tests and fired in accordance with table I.

4.2.2.2 Vibration. Cartridges shall be subjected to the vibration test using Drawing 844AS291 (see 4.2.2.2.1). After the vibration test has been completed, the cartridges shall be subjected to the nondestructive tests and fired in accordance with table I.

4.2.2.2.1 Procedure.

a. Verify that the vibration test fixture does not have any resonant vibration frequency between 5 and 2000 Hz (see figure 1).

b. Secure cartridges into the vibration test fixture using a torque of 45 ± 3 inch-pounds. Antiseize compound shall be applied to the fixture threads prior to installation of the cartridges. The three mutually perpendicular axes shall be used for vibration testing. Vibration time shall be one hour per axis per temperature for a total of nine hours.

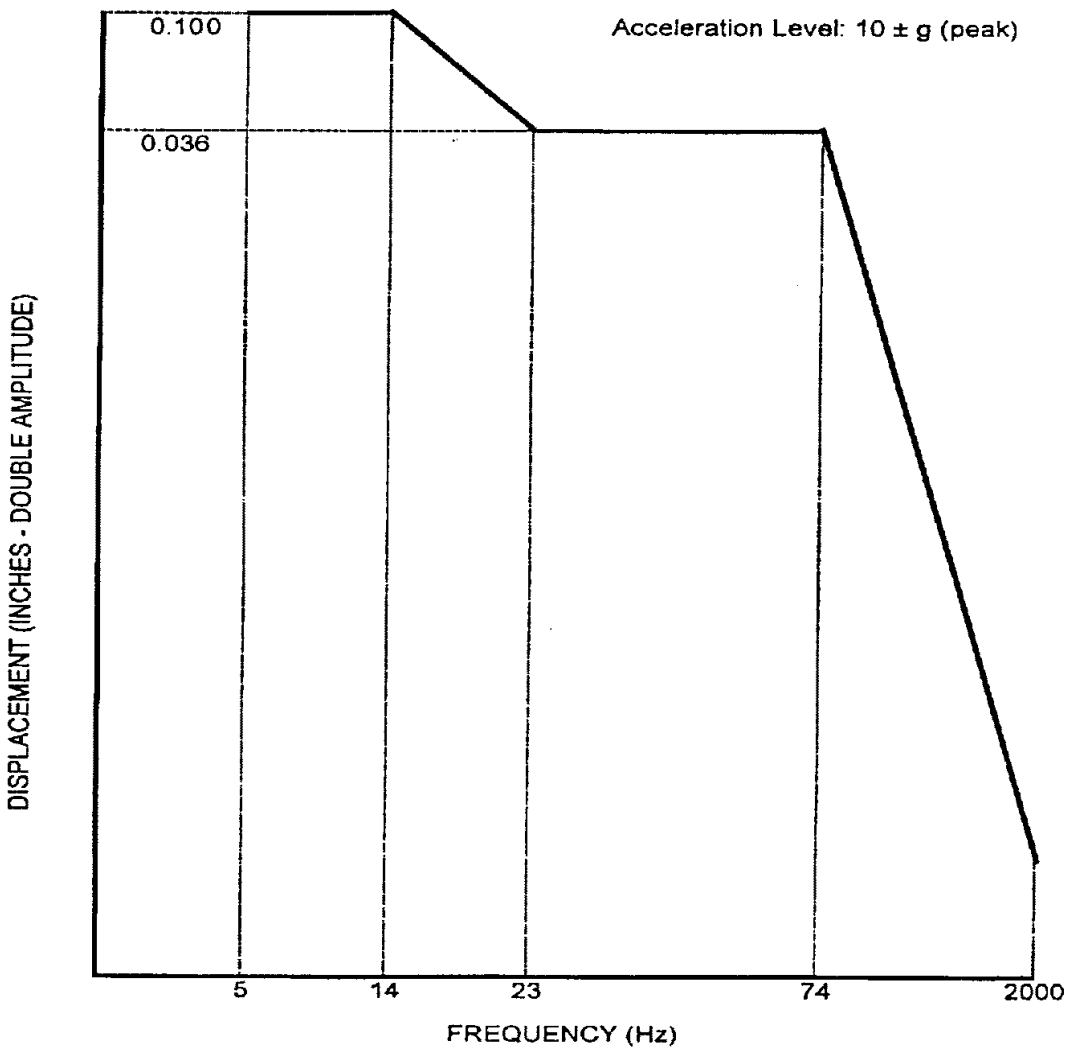


FIGURE 1. Vibration H-curve.

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c. Perform sinusoidal vibration cycling at 70°F for one hour with the frequency varying logarithmically from 5 to 2000 Hz and return to 5 Hz in approximately 20 minutes to the levels. Continue this until one hour of cycling has been completed.

d. After completion of the one-hour cycling period, the vibration fixture shall be rotated 90 degrees so that the cartridges are positioned in a different axis than in subparagraph c. above. The cartridges in this new axis position shall then be vibrated per subparagraph c. above for one hour at 70°F.

e. After completion of the second one-hour cycling period, the vibration fixture shall again be rotated 90 degrees so that the cartridges are positioned in a different axis than in subparagraphs c. or d. above. The cartridges in this new axis position shall then be vibrated per subparagraph c. above for one hour at 70°F.

f. After completion of three one-hour cycling periods (one hour per axis) at 70°F, the procedure outlined in subparagraphs c. through e. above shall be repeated with the temperature at 160°F in lieu of 70°F.

g. After completion of three one-hour cycling periods (one hour per axis) at 160°F, the procedure outlined in subparagraphs c. through e. above shall be repeated with the temperature at -65°F in lieu of 160°F.

4.2.2.3 Temperature/humidity/altitude cycling. Cartridges shall be subjected to the temperature/ humidity/altitude cycling test as outlined below. Provision has been for withdrawal of cartridges at two times during the deterioration, if such exists. The schedule has been arranged in a manner that operations are not required outside normal working hours except for such supervision to ensure proper operation of the test equipment. It is not mandatory that the day/clock time schedule given below be followed; however, it is mandatory that the time, environmental, and sequence requirements be followed. A minimum of two chambers are needed to conduct the temperature shock portion of this test. If it is desired to use three chambers, it is permissible provided the time, environmental, and sequence requirements are met. The fluctuations from the specified temperatures shall not exceed 5°F. When the temperature chamber door is opened to place test items inside, the chamber pressure will be atmospheric. The time required to return the chamber pressure to 0.65 pounds per square inch absolute (psia) shall not exceed one hour. Cartridges shall be supported on screen trays or racks so that all areas are exposed to the same prescribed atmospheric conditions at all times throughout the test. The schedule to be followed is:

Monday	0800	Place test items in a chamber maintained at 70°F at 50 percent (%) relative humidity (RH).
	1200	Raise chamber temperature to 160°F and the RH to 95%. The chamber temperature shall reach 160°F at 95% RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -65°F and adjust for a pressure altitude of 70,000 feet (0.65 psi).
Tuesday	0800	Remove test items from above chamber and place test items in a chamber maintained at 70°F at 50% RH.
	1200	Remove test items from above chamber and immediately place in a chamber

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		maintained at -65°F and adjust for a pressure altitude of 70,000 feet (0.65 psi).
	1600	Remove test items from above chamber and immediately place in a chamber maintained at 160°F at 95% RH.
Wednesday	0800	Reduce chamber pressure to 70°F at 50% RH. The chamber temperature shall reach 70°F at 50% RH not later than 0900.
	1200	Raise chamber pressure to 160°F and 95% RH. The chamber temperature shall reach 160°F at 95% RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -65°F at 95% RH and adjust for a pressure altitude of 70,000 feet (0.65 psi).
Thursday	0800	Place test items in a chamber maintained at 70°F at 50% RH.
	1200	Remove test items from above chamber and immediately place in a chamber maintained at -65°F and adjust for a pressure altitude of 70,000 feet (0.65 psi).
	1600	Remove test items from above chamber and immediately place in a chamber maintained at 160°F at 95% RH.
Friday	0800	Reduce chamber pressure to 70°F at 50% RH. The chamber temperature shall reach 70°F at 50% RH not later than 0900.
	1200	Raise chamber pressure to 160°F and the RH to 95%. The chamber temperature shall reach 160°F at 95% RH not later than 1300.
	1600	Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a standard ambient pressure.

This schedule shall be followed for a total of four weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of 160°F at 95% RH. Six cartridges each shall be removed at 0800 on the third and fourth Monday. The completion of the test occurs on the fifth Monday after starting. After the temperature/humidity/altitude test has been completed, the cartridges shall be subjected to the nondestructive tests and test fired in accordance with table I. Failure of any cartridge to meet the requirements of 3.7.3 shall cause rejection of the first article.

4.2.3 Functional test sampling.

4.2.3.1 Low temperature (-65°F). Test devices and cartridges shall be conditioned at a temperature of -65°F for a minimum of 6 hours after the temperature conditioning chamber reaches thermal equilibrium and then fired. Ballistic testing shall be in accordance with table I. Failure of any cartridge to meet the requirements of 3.9 shall cause rejection of the first article.

4.2.3.2 Normal temperature (70°F). Test devices and cartridges shall be conditioned at a temperature of 70°F for a minimum of 6 hours after the temperature conditioning chamber reaches thermal equilibrium and then fired. Ballistic testing shall be in accordance with table I. Failure of any cartridge to meet the requirements of 3.9 shall cause rejection of the first article.

4.2.3.3 High temperature (160°F). Test devices and cartridges shall be conditioned at a temperature of 160°F for a minimum of 6 hours after the temperature conditioning chamber reaches thermal equilibrium and then fired. Ballistic testing shall be in accordance with table I. Failure of any cartridge to meet the requirements of 3.9 shall cause rejection of the first article.

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4.3 Production inspection. Unless otherwise specified, sampling procedures and rejection criteria, except for the ballistic tests of 4.8, shall be in accordance with the requirements of ANSI/ASQC Z1.4. Critical, major, and minor characteristics shall be inspected for defects to AQLs as specified in the contract (see 6.2 and 6.7). The requirements of ANSI/ASQC Z1.4 pertaining to normal and tightened inspections shall apply. A production lot of cartridges shall contain main propellant charges and primers from single production lots only. The minimum and maximum production lot size shall be 181 and 800 cartridges, respectively.

4.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major or minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.5 Quality conformance. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL825AS100 and the verification of the requirements listed in table II.

TABLE II. Quality conformance inspection.

Item	Nature of requirement verification	Applicable para.	Classification
Cartridge	Radiographic	4.7.3	C1
Cartridge	Leakage	4.7.4	C2
Primary component	Single lot	3.3.1	M101
Cartridge	Visual inspection	4.7.2	M102
Cartridge	Packaging & container marking	3.10, 5.1	M103

4.6 Lot acceptance inspections. Lot acceptance inspection shall consist of the examinations and tests specified in table III. Failure of any sample cartridge to comply with the requirements listed in table III shall be cause for rejection of the lot represented.

TABLE III. Ballistic test for production sample.

Nature of requirement	Sample		Para.
Visual inspection	50	*	4.7.2
Radiographic examination	50	*	4.7.3
Leakage	50	*	4.7.4
Ballistic test @ 70 ± 5°F	50		4.8

* Retain samples, see contract

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4.7 Tests.

4.7.1 Conformance to drawings. Each cartridge of the first article and the production sample shall be examined for conformance to the drawing requirements of DL825AS100. Failure of the cartridges to meet the requirements of the drawings shall cause rejection.

4.7.2 Visual inspection. Examine cartridges of the first article and the production sample and reject those having any of the defects of 3.2 and 3.4.

4.7.3 Radiographic examination. All cartridges shall be radiographically examined in accordance with ASTM E 1742 for conformance to the assembly requirements of 3.5. The cartridge shall be in a vertical position with the longitudinal axis of the powder case perpendicular to the source of radiation and with the black dot on the primer housing facing the source of radiation. Tap the cartridge case lightly with a small nonmetallic object to level the propellant charge. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number one (1) in each production lot. The cartridge shall be arranged on trays or boards in consecutive numerical order and each radiograph shall carry a permanent identification of the cartridge displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the x-ray. The serial number shall be identified on the x-ray. The serial numbers shall be ink stamped on the cartridge correctly and legibly. Radiographs of the entire lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective cartridges shall be annotated on the vendor's report and discarded.

4.7.4 Leakage. Each cartridge of the first article and the production sample shall be tested with a dry gas leak detector of sufficient sensitivity to verify conformance with the requirements of 3.6. Failure of any cartridge of the production sample to meet the requirements of 3.6 shall be cause for each cartridge of the production lot to be tested. Any cartridge of the production lot failing to meet the requirements of 3.6 shall be rejected.

4.8. Ballistic test.

4.8.1 Test procedure. With the exception of the cartridges tested in association with the functional tests of 4.2.3, the cartridges to be tested shall be conditioned for not less than 6 hours at 70 ± 5°F. They shall then be fired in a ballistic test fixture conforming to Drawing 838AS144. Two slugs weighing 2.50 ± 0.05 ounces each shall be ejected in opposite directions from each other in each test. Failure of a first article test sample to meet the requirements of 3.9.1 or any cartridge of a first article test sample to meet the requirement of 3.9.2 shall cause the first article to be rejected.

4.8.2 Data. Data to be reported shall include:

a. Time from sear removal to first indication of motion for each slug ejected.

b. Average velocity and time required for each slug to traverse a distance of 36.000 ± 0.125 inches from its original position. Timing shall begin at first indication of slug motion.

4.8.3 Retest. There shall be no retests.

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4.9 Humidity. The relative humidity of the working chamber shall be monitored by test equipment of appropriate accuracy. All cartridges manufactured during the period of time that the requirements of 3.3.2.1 were not met shall be rejected.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The CCU-33/A impulse cartridge provides a power source for the rapid deployment of parachutes. This is accomplished by firing 14 slugs (2.5 ounces each) which are attached to the hem of the parachute, from a Stencel Aero Engineering Corporation Spreading Gun Assembly, Stencel Part Number SK 86-0051-1. The CCU-33/A impulse cartridge is military unique, as it is used for aircrew escape purposes and thus has no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).

c. Whether a first article sample is required (see 3.1 and 4.2).

d. Packaging (see 3.10 and 5.1).

e. Whether samples are to be retained (see 4.2 and 4.6).

f. Production lot size and test activity (see 4.3).

g. Directions for shipping radiographs of entire production lot along with the ballistic sample to the activity conducting the first article and production lot tests.

h. Marking, if other than as specified (see 6.5).

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i. That the safety precaution requirements of the "DoD Contractors' Safety Manual for Ammunition and Explosives," DoD 4145.26-M, are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.

j. Whether AQLs must be specified (see 4.2.1, 4.3, 6.6, and 6.7).

k. Whether data is required (see 4.8.2).

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition. For the purpose of this specification, the following definitions apply.

6.4.1 Functioning time. Functioning time is the time interval between removal of the sear from the firing mechanism to the first indication of slug motion.

6.4.2 Loose crimp. A crimp which allows relative movement when twisted by light hand torque.

6.4.3 Cracks. A complete separation of material.

6.4.4 Fissures. The surface separation of material.

6.5 Marking.

6.5.1 Special marking. Marking of inner and outer containers shall be in accordance with the applicable drawings listed on DL844AS275 and with 49 CFR 171-199.

6.5.2 Normal marking. Unless otherwise specified in the contract or order, the marking information of the inner and outer containers shall be as specified below:

Inner Container Marking

a. NSN/DODIC 1377-01-119-8553/MF78
b. Cartridge, Impulse, CCU-33/A
c. Drawing 825AS100
d. Quantity

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e. Lot number in accordance with MIL-STD-1168
f. DOT explosive Class C

Outer Container Marking

a. NSN/DODIC 1377-01-119-8553/MF78
b. Cartridge, Impulse, CCU-33/A
c. Drawing 825AS100
d. Quantity
e. Gross weight and cube
f. Lot number in accordance with MIL-STD-1168
g. Contract or purchase order number
h. WARNING: EXPLOSIVE DEVICE, CLASS C EXPLOSIVE, HANDLE CAREFULLY, KEEP AWAY FROM FIRE

Additionally, the above specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. 6 First article sampling. The following AQLs have been used successfully for first article sampling. The contracting activity is cautioned that any deviation for these ANSI/ASQC Z1.4 selections should be warranted and verified statistically considering the criticality of the cartridge's function.

a. For minor characteristics, an AQL of 2.5% has been used

b. For major characteristics, an AQL of 1.0% has been used

c. For critical characteristics, an AQL of 0.065% has been used

6. 7 Production inspection sampling. The following AQLs have been used successfully for production inspection sampling. The contracting activity is cautioned that any deviation for these ANSI/ASQC Z1.4 selections should be warranted and verified statistically considering the criticality of the cartridge's function.

a. For minor characteristics, an AQL of 2.5% has been used

b. For major characteristics, an AQL of 1.0% has been used

c. For critical characteristics, an AQL of 0.065% has been used

Preparing activity:
Navy - OS
(Project 1377-0077)

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